TEXORA · LECTURE SHEET

Surface Decoration Technique

Ornament applied to the fabric surface — printing, embroidery, appliqué, quilting, smocking, dye-resist patterning and embellishment — classified by how the decoration is made, with the production logic of each.

AM-4203 · FASHION DESIGN AND PRODUCT ENGINEERING · TOPIC 3 · LEVEL 4 TERM 2

1. What counts as surface decoration

Surface decoration is pattern and texture created **on** a finished fabric, as distinct from pattern built *into* the fabric by weaving or knitting structure (the subject of TE-3205). Because it is applied after the cloth exists, it can be placed exactly where the design wants it — a motif on one pocket, a border at one hem — and it can be added at garment stage as well as at fabric stage. The designer's classification is by mechanism, because the mechanism decides cost, durability and hand-feel.

2. The techniques, by mechanism

FAMILY	TECHNIQUES	HOW IT WORKS AND WHERE IT IS USED
Printing (colour applied through a pattern)	Block, screen (flat and rotary), roller, transfer, digital inkjet	Colour paste or ink passes to the cloth only where the pattern allows. Block: a carved block is inked and pressed by hand — slow, artisanal. Screen: paste is forced by squeegee through a mesh stencil, one screen per colour. Digital: inkjet heads print the file directly — no screens, economical for short runs and photographs. (Industrial print processes in depth: TE-3105.)
Embroidery (pattern built with thread)	Hand embroidery stitches; machine embroidery (AM-4202 Study 2)	Thread is stitched through the ground fabric to build motifs with a raised, textured surface. Running, satin and fill stitches form the vocabulary; density and direction of stitch create shading no flat print can imitate.
Appliqué (fabric added to fabric)	Raw-edge, turned-edge, reverse appliqué	A shaped piece of a second fabric is laid on the ground and stitched down at its edges; in reverse appliqué the top layer is cut away to reveal the layer beneath. Gives bold, flat shapes with a textile — not printed — edge.

Quilting (layers stitched together)	Hand and machine quilting, trapunto	Two (or three, with wadding) layers are stitched through in patterns; the stitching line itself becomes the design and the raised areas between lines give the surface its relief.
Manipulated surface (the fabric itself is re-formed)	Smocking, shirring, tucks, pleats, ruching	Stitching gathers or folds the ground fabric into a permanent textured pattern. Smocking: fabric is gathered into regular pleats and stitched over them with decorative stitches, giving controlled stretch — the traditional decoration of children's dresses (AM-4202 Study 3's category). Tucks and pleats fold the cloth into ridges that are stitched down.
Resist and dye patterning	Tie-dye, batik, shibori, ikat (yarn-stage)	Parts of the cloth are protected — tied, waxed, clamped, stitched — before dyeing, so the pattern appears as undyed (or differently dyed) areas. Batik's wax resist and tie-dye's bound resist produce the soft, organic edges no printing screen reproduces exactly.
Embellishment (objects attached)	Sequins, beads, mirrors, studs, lace application, fringe	Separate elements are sewn or fixed onto the surface. Highest visual value per area and the most fragile in wear and washing — placement must respect both the design and the care label.

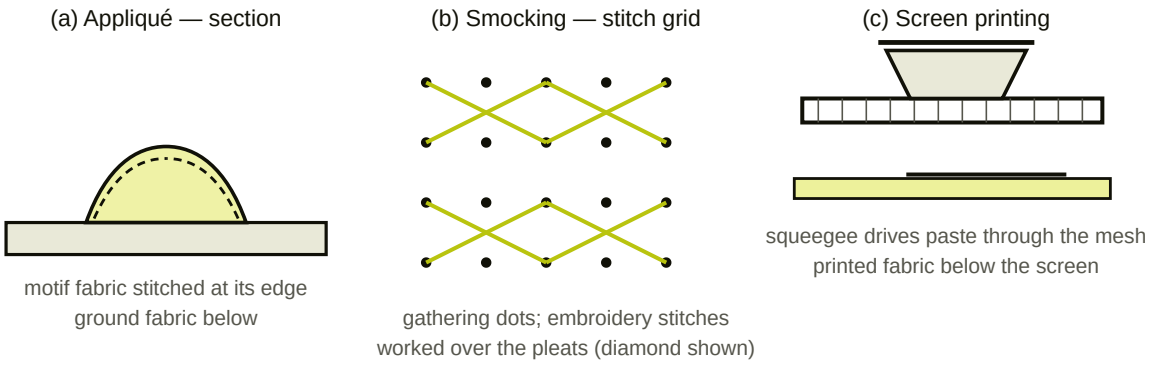


Figure 1 — Three mechanisms, three sections: appliqué adds a layer, smocking re-forms the layer itself, screen printing adds colour through a stencil. The mechanism — not the motif — is the classification.

3. Choosing a technique: the design trade-offs

QUESTION	HOW THE TECHNIQUES ANSWER
Relief and texture?	Embroidery, quilting, smocking and embellishment give real three-dimensional surface; printing and resist dyeing stay flat.

Fine detail and shading?	Digital printing reproduces photographic detail; embroidery shades by stitch direction and density; appliqué gives bold shapes only.
Cost behaviour with quantity	Screen printing has high set-up (a screen per colour) and low unit cost — it rewards long runs. Digital printing has almost no set-up and a steady unit cost — it rewards short runs. Hand techniques (block, hand embroidery, batik) keep a high labour cost in every piece at any quantity.
Durability in wear and wash	Woven-in effects outlast everything; prints and embroidery are durable when properly fixed/executed; attached embellishment is the most vulnerable and drives the care instructions.
Where it can be applied	Fabric-stage decoration suits all-over patterns; garment-stage decoration (embroidery, appliqué, placement print) puts the motif exactly on a finished part — at the price of handling garments instead of cloth.

4. Designing with surface decoration

Three working rules govern good use. **Scale:** the motif must be sized to the garment panel it will occupy — a print designed for a dress front fails when cut into a child's sleeve. **Placement:** decoration is positioned against the body's landmarks (Topic 2's division lines) — at the neckline, the waist, the hem — because the eye reads the garment from those points. **Restraint:** one dominant technique per design area; surface decoration competes with silhouette (Topic 4), and when both shout, neither is heard. The mood board (Topic 6) is where these choices are first tested together.

Glossary

TERM	MEANING
Appliqué	Decoration made by stitching a shaped fabric piece onto a ground fabric.
Smocking	Gathering fabric into regular pleats and embroidering over them, producing a decorative, elastic surface.
Resist dyeing	Patterning by protecting parts of the cloth (wax, ties, clamps, stitches) from the dye.
Trapunto	Quilting in which selected areas are additionally stuffed from behind to raise them in relief.
Placement print	A print positioned on one defined part of a garment rather than repeated all over.

Key points and summary

- Surface decoration is applied on finished cloth and classified by mechanism: printing, embroidery, appliqué, quilting, manipulation, resist dyeing, embellishment.
- Mechanism decides relief, detail, cost behaviour with quantity, durability and stage of application.
- Screen printing rewards long runs, digital rewards short runs, hand techniques carry labour cost in every piece.
- Design use is governed by scale, placement against body landmarks, and restraint.

Practice

Answer in your own words; brief model answers follow.

1. Classify by mechanism: (a) wax-patterned cloth dyed indigo; (b) a motif of cord stitched through two layers with wadding; (c) sequins sewn along a neckline; (d) a child's bodice gathered into stitched diamond pleats.
2. A designer needs 300 garments with a photographic chest motif, and later possibly 30,000 with a two-colour all-over pattern. Recommend a printing route for each and justify by cost behaviour.
3. Why is embellishment placed with the care label in mind?
4. State the three design rules of §4 and apply one to a stated example of your own.

Answer key

1. (a) Resist dyeing (batik family). (b) Quilting. (c) Embellishment. (d) Manipulated surface — smocking.
2. 300 pieces: digital printing — near-zero set-up and photographic reproduction; screen set-up for many colours cannot be recovered at this quantity. 30,000 pieces: screen (rotary) printing — screens are a fixed cost spread over the run, unit cost falls below digital, and two colours need only two screens.
3. Because attached elements are the most wash- and wear-sensitive part of the garment: their durability sets the washing instructions, and decoration that cannot survive the garment's normal care destroys the garment's value first.
4. Scale (motif sized to its panel), placement (against body landmarks), restraint (one dominant technique per area). Example credited for any correct application, e.g. a single embroidered motif at the neckline rather than embroidery, print and sequins competing across one bodice.